

The Pop-Up Metaverse: A Multi-User, Multi-Tasking Spatial Computing Environment for Collaborative Spatial Problem-Solving.

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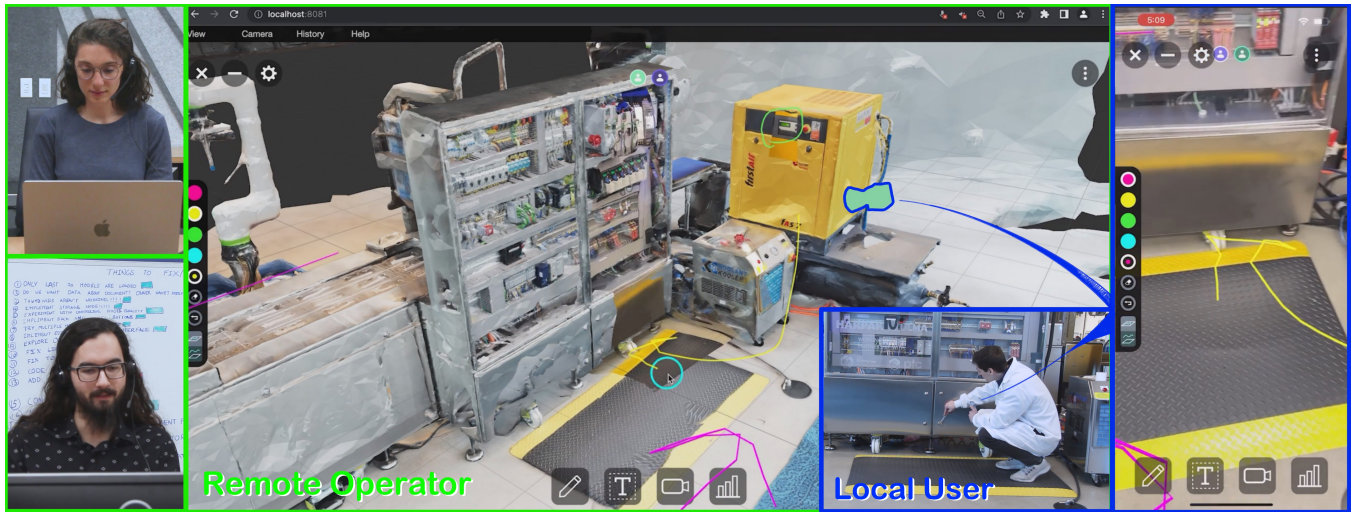


Figure 1: The *Pop-Up Metaverse* Workflow: Remote operators collaborate seamlessly via web browsers (left), guiding local users (right) equipped with iPhones, showcasing web-based, installation-free spatial problem-solving.

ABSTRACT

We introduce the *Pop-Up Metaverse*, an innovative web-based system seamlessly blending spatial computing with 3D visualization to catalyze collaborative problem-solving in the Industrial Metaverse. The workflow commences with a local user swiftly scanning the environment using a LiDAR-equipped iPhone/iPad, generating a baseline mesh model. Elevating visual fidelity, the system employs the state-of-the-art 3D reconstruction technique, Gaussian-splatting,

for captivating photorealistic effects. This enriched 3D canvas becomes a hub for multi-user collaboration, emphasizing the Pop-Up nature facilitated by web technology—no installations required. Key features encompass Remote Assistance, enabling real-time expert guidance within the 3D scene; Human Motion Recorder, scrutinizing movements and postures in recorded human activity; and Spatial Analytics, meticulously tracking environmental changes over time. Ensuring accessibility, remote participation mandates only a web browser, significantly reducing entry barriers to the metaverse experience. The *Pop-Up Metaverse* empowers a workflow tailored for solving spatial problems by seamlessly uniting the physical and digital realms.

CCS CONCEPTS

- Human-centered computing → User interface toolkits; Visualization toolkits; Collaborative and social computing systems and tools; Systems and tools for interaction design.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3648677>

KEYWORDS

Pop-Up Metaverse, Spatial Computing, Remote Collaboration

ACM Reference Format:

Benjamin Reynolds, James Hobin, Yuanzhi Cao, Martin Klaudiny, Daniel Dangond, Kaize Xie, Keranie Theodosiou, Rik de Leeuw, Shiva Kashalkar, and Valentin Heun. 2024. The Pop-Up Metaverse: A Multi-User, Multi-Tasking Spatial Computing Environment for Collaborative Spatial Problem-Solving. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3648677>

1 INTRODUCTION

As we embark on the transformative journey of Industry 4.0 [6], where technologies like Spatial Computing, Artificial Intelligence (AI), Internet of Things (IoT), and advanced data analytics are reshaping industrial landscapes, the emergence of the *Pop-Up Metaverse* signifies a pivotal shift towards a more accessible and collaborative spatial computing environment.

In the realm of Industry 4.0, Spatial Computing stands as a beacon, defining the convergence of the physical and digital worlds to unlock novel forms of interaction [12]. Its transformative potential spans various industrial sectors, promising revolutionary changes in assembly lines, maintenance, and training through immersive experiences that accelerate learning and enhance procedural retention [5]. However, the practical deployment of Spatial Computing in industry encounters challenges associated with sophisticated, high-end equipment, high costs, and intricate calibration requirements [15].

The *Pop-Up Metaverse* sets itself apart by providing a user-friendly and effortlessly accessible tool directly through a standard web browser, eliminating the need for cumbersome installations. The metaverse experience commences as a local user swiftly performs an environmental scan using a LiDAR-equipped iPhone/iPad, creating a textured mesh model for the initial 3D visualization. Subsequently, the data seamlessly undergoes transformation into a photorealistic 3D scene in the backend cloud server, achieved through the efficient 3D reconstruction using 3D Gaussian-splatting [8]. This streamlined and straightforward approach proves particularly advantageous in dynamic industrial settings like factories, where the necessity for intricate preparations is completely eliminated.

The moniker "Pop-Up Metaverse" captures the essence of the system's nature and purpose. "Pop-Up" reflects the ease with which this metaverse is initiated—a quick scan with an iOS device opens the door to collaboration, with remote users seamlessly joining via a browser link, devoid of intricate setups. This simplicity aligns with the demands of industrial environments, where time is of the essence, and complex preparations are impractical.

The term "Metaverse" encapsulates the system's ubiquity and inclusivity [10]. Our spatial computing environment encompasses a myriad of technologies, including Augmented and Virtual Reality, 3D scanning and capture, spatial video recording, human pose detection, and more. In essence, it is a comprehensive space where state-of-the-art technologies seamlessly coexist. Leveraging web and cloud-centric infrastructure akin to the internet, our metaverse operates on the principle of connectivity through a simple link. This flexible and scalable environment can seamlessly incorporate

cutting-edge web APIs, such as ChatGPT [11], further enhancing its collaborative spatial problem-solving capabilities, especially in complex industrial settings. The *Pop-Up Metaverse* is not merely a platform; it is a dynamic, adaptable, and inclusive spatial computing environment tailored for the collaborative resolution of spatial challenges in the industrial landscape. The *contributions* of this paper are described as follows:

- The innovative workflow design, seamlessly progressing from local environmental scanning to web browser accessible collaborative problem-solving.
- The design of a cloud and web-based data infrastructure that facilitates an immersive 3D spatial computing experience within the *Pop-Up Metaverse*.
- The detailed design and implementation of interaction/interface features within the *Pop-Up Metaverse*, offering users efficient tools for effective spatial collaboration, including its core features such as remote assistance, human motion analysis, and spatial analytics.

2 THE POP-UP METAVERSE SYSTEM OVERVIEW

2.1 System Workflow Design

The *Pop-Up Metaverse* initiates with a local user effortlessly conducting an environment scan using an iPhone/iPad equipped with LiDAR technology, enabling both scanning and precise localization/tracking of the mobile device for AR delivery to the local user using PTC's Vuforia [4]. This swift process, taking just a few minutes, generates a 3D textured mesh of the scanned area, immediately accessible to remote users for immersive 3D visualization via a simple web browser link. This streamlined scanning approach is particularly advantageous in industrial contexts, addressing potential process interruptions, such as failing sensors and motors, in high-value throughput environments.

Within this collaborative 3D space, a multitude of local users equipped with iPhones and iPads, along with remote users accessing via web browsers, seamlessly collaborate (Figure 1). The platform's capabilities extend to embedding and spatial contextualization of documents, recording and streaming spatial videos, and incorporating interactive drawings and instructions within the digital environment. These features collectively create a dynamic and interactive representation of the physical world, uniquely suited for addressing immediate on-site challenges. The design of the *Pop-Up Metaverse* places a strong emphasis on accessibility, requiring local users only to have an iPhone or iPad, and the application, while remote users can effortlessly engage through a straightforward browser link, eliminating the need for specialized hardware. This user-friendly approach ensures the *Pop-Up Metaverse* is readily adoptable in industrial settings, enabling quick and effective collaborative problem-solving.

2.2 Web and Cloud Integration

The *Pop-Up Metaverse* integrates its functionalities within a dual-layered infrastructure, seamlessly blending local edge computing with cloud server connectivity. Operating a Node.js server locally on the iOS device, the system captures scan data and initiates sessions

within the phone itself. Initially designed for edge functionality without cloud dependence, this approach enables local users to initiate sessions even in environments with limited or no internet access.

The local server on the iPhone, housing an edge-agent module, establishes communication with an Azure-based cloud proxy server [2]. Upon scanning a new area, the local server acquires a unique network ID through the edge-agent, enabling session identification on the cloud proxy server. An invite link, incorporating the network ID as a path on the proxy server, is generated. HTTP requests to this link are then seamlessly proxied to the local phone's server through the cloud proxy server.

The system employs real-time-focused web technologies, utilizing HTTP and WebSockets paired with a reverse proxy, ensuring a low-latency, seamless web-browser-based experience. The focus on tools inherent to major browsers enables our system to reach a large number of users across diverse operating systems and device form factors.

A fundamental innovation lies in the infrastructure supporting RGB+Depth video streaming. Leveraging peer-to-peer WebRTC connections enhances privacy and reduces hosting costs. Introducing a new protocol for each video and depth frame, with precise timing and camera pose data, enables remote operators to play back and position RGB+D video precisely within the static scan. Additionally, a WebAssembly implementation of the open-sourced RVL depth encoding technique [14], tailored to iPhone depth sensor data characteristics, significantly reduces the size and compression time of each depth frame while maintaining compatibility with modern web browsers. The peer-to-peer WebRTC connection for transferring frames of our RGB+Depth protocol provides a unique augmented real-time view of spatial context on any device with a web browser. This innovative integration ensures the *Pop-Up Metaverse* operates flexibly in either an "online" cloud-connected mode or an "offline" edge mode, accommodating diverse industrial scenarios and offering an accessible experience to users across devices.

2.3 Advanced 3D Visualization with Gaussian-Splatting

In industrial settings, achieving high visual quality in 3D models is crucial, particularly when replacing physical visits with virtual environments in applications like remote assistance. While the *Pop-Up Metaverse* excels in rapid mesh model generation, certain materials and surfaces pose challenges, such as reflective surfaces, large uniform regions, and transparent materials like glass [13].

To overcome these challenges, the *Pop-Up Metaverse* incorporates advanced 3D visualization techniques. While Neural Radiance Fields (NeRF) offer high-quality renderings [9], their resource-intensive nature makes them less practical for real-time use. As an alternative, Gaussian-splatting proves effective [8]. Trained splat models, processed on a cloud server (specifically Microsoft Azure), enable efficient rendering in a web browser at high frame rates (60+ fps) using WebGL technology [1].

Our implementation utilizes the original Gaussian-splatting method to train splat models on the cloud. When scan data is uploaded, it undergoes processing and training, taking variable

time depending on the scanned area's size. Within approximately 20 minutes, the trained model, with a file size below 100 MB for a regular room-size scan, is ready and sent back to the server. This allows remote users to experience a photorealistic, real-time 3D visualization in their web browsers, significantly enhancing realism and detail compared to traditional mesh models. A comparative illustration in Figure 2 demonstrate the superior rendering effects of Gaussian-splatting, highlighting the substantial improvement in visual fidelity.

3 FEATURED APPLICATIONS

3.1 Spatial Remote Assistance

A cornerstone application of the *Pop-Up Metaverse* is Remote Assistance, meticulously crafted to foster efficient collaboration between remote experts and on-site users in industrial settings. Particularly instrumental in tasks like machine repair, this feature empowers remote experts to guide local users through intricate procedures. Leveraging the *Pop-Up Metaverse*, remote users seamlessly access the local user's first or third-person camera views, offering precise focus on specific areas or components within the space. Spatial pointers (Figure 3-(1)) and drawing tools (Figure 3-(2)) facilitate real-time bi-directional interaction, allowing experts to highlight areas of interest or urgency and create visual cues within the environment. The spatial video tool (Figure 3-(3)) enhances remote visualization through live streaming or recording of 3D videos. Furthermore, the system enables experts to search and integrate relevant documentation, such as PDFs, directly into the scene for the local user's reference. Enhanced communication features include opening a chatbox for detailed repair logs or instructions. Notably, the *Pop-Up Metaverse* supports multi-to-multi collaboration, enabling several remote experts to concurrently assist local users. This collective diagnostic and problem-solving capability positions the *Pop-Up Metaverse* as an indispensable tool in complex industrial operations, streamlining processes and elevating the efficiency and accuracy of maintenance and repair tasks.

3.2 Human Motion Recorder

The second core feature is the "Human Motion Recorder," an application crafted for capturing and analyzing human activity in industrial environments. With iPhones and iPads, this feature allows the recording of 3D spatial videos capturing individuals performing tasks, such as operating machinery. Unlike conventional visual captures, these recordings are immersive 3D videos with color and depth, spatially anchored in the environment. Going beyond visual documentation, the *Pop-Up Metaverse* utilizes a mobile-friendly AI model for real-time 3D human pose estimation [3]. This model dynamically infers a 3D skeleton pose from live camera frames, accurately placing it into the scan using Lidar depth (Figure 4-(1)). The system incorporates a spatial analyzer tool to record and analyze these 3D human poses over time and space, providing insights for assessing factory floor efficiency, identifying safety concerns, and optimizing workflow (Figure 4-(2)). Additionally, the analyzer computes a REBA (Rapid Entire Body Assessment) score [7], evaluating human poses with color-coded highlights to pinpoint potentially straining body postures (Figure 4-(3)). This comprehensive analysis

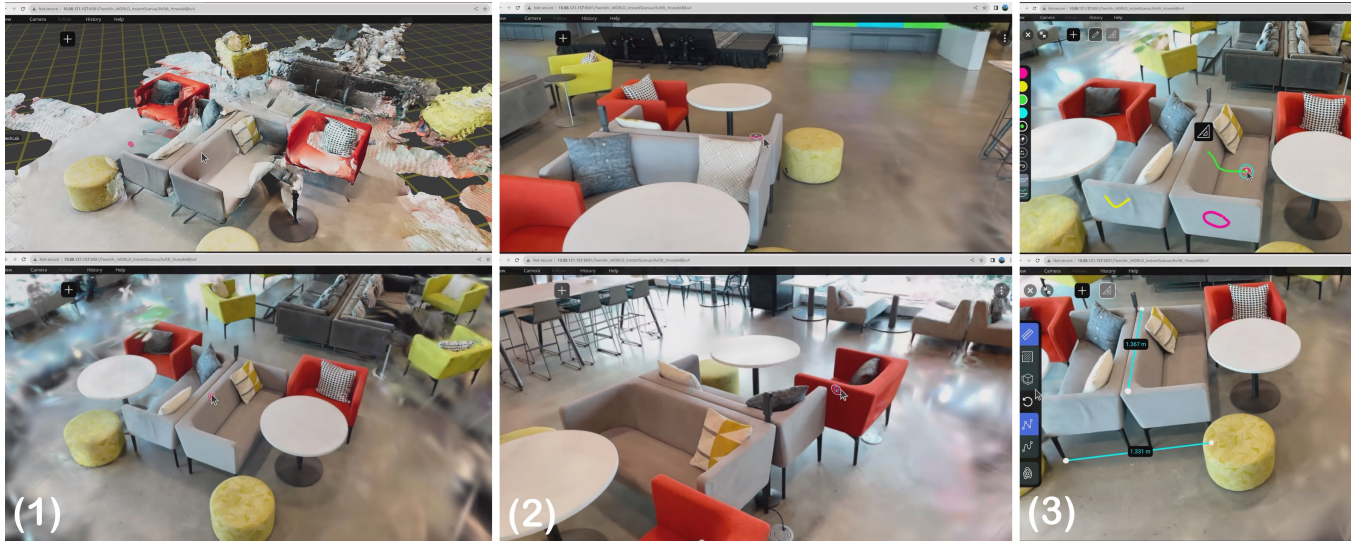


Figure 2: Real-time 3D Visualization with Gaussian-Splatting. (1-top) Lidar mesh scan vs. (1-bottom) Gaussian-Splatting: enhanced rendering capturing details like a featureless table. (2) Various perspectives demonstrate consistent superior quality. (3) Drawing and measurement tools showcase interactive collaboration in the *Pop-Up Metaverse*.



Figure 3: Example Tools for Spatial Remote Assistance. (1) The Spatial Pointer directs attention with highlighted projections, aiding in precise guidance. (2) The Spatial Drawing Tool empowers users to mark targets or draw paths for collaborative problem-solving. (3) The Spatial Video Tool offers live streaming and recording of spatial RGB+D videos, providing spatial documentation of events.

is vital for addressing ergonomic concerns and prioritizing workforce well-being. The "Human Motion Recorder" feature emerges as a potent tool, reshaping the landscape of operational efficiency and safety in industrial settings by transforming the monitoring, analysis, and optimization of human activity.

3.3 Spatial Analytics

The *Pop-Up Metaverse* culminates its core features with "Spatial Analytics," a robust tool meticulously designed to monitor and analyze changes within industrial environments. This tool leverages the system's capacity to record and compare the current in-camera scene with previous scans, utilizing a red/green voxels for visualizing missing/added objects (Figure 5-(1)). In intricate industrial settings, where tracking environmental changes is paramount, the "Spatial Analytics" tool proves indispensable. It excels in identifying relocated items, diagnosing issues, and proactively addressing potential problems by providing a clear and immediate visual representation of alterations in the workspace. Additionally, the tool incorporates

a measurement feature for in-situ distance/area measurements and a spatial map for a comprehensive global visualization of depth and gradient (Figure 5-(2)). As a final touch, ChatGPT integration within our metaverse experience adds a unique layer of functionality, enabling not only general question answering but also accessing and performing tasks, such as spatial tool searching and filtering, within the *Pop-Up Metaverse* (Figure 5-(3)).

4 CONCLUSION

The *Pop-Up Metaverse* signifies a revolutionary stride in integrating spatial computing into industrial settings, aligning seamlessly with the principles of Industry 4.0. Its streamlined workflow, starting from LiDAR-based scanning with a mobile device to creating a collaborative 3D environment accessible via web browsers, facilitates the adoption of spatial computing in industry. The core features—Spatial Remote Assistance, Human Motion Recorder, and Spatial Analytics—collectively amplify efficiency, safety, and collaborative problem-solving. Positioned at the forefront of technological



Figure 4: Human Motion Recorder. (1) Records the skeletal motion of a human worker. (2) Captures the spatial trajectory over time and space. (3) Allows segmentation into individual actions for in-depth analysis, including REBA score for evaluating ergonomics.

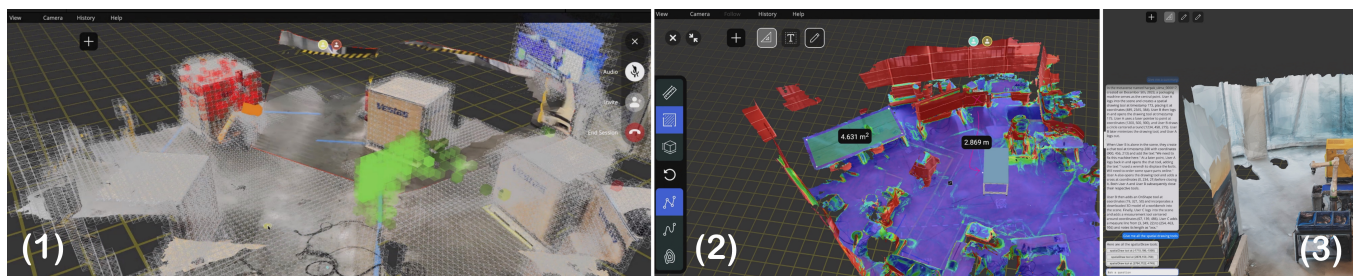


Figure 5: Spatial Analytics feature. (1) Detection of spatial anomalies using a green/red voxels. (2) In-situ measurement showcased on a gradient color map, where blue color marks navigatable regions. (3) ChatGPT integration for spatial tool search.

advancement, the *Pop-Up Metaverse* not only refines current industrial practices but also lays the foundation for future innovations. It represents a pivotal step toward realizing the potential of intelligent, interconnected industrial ecosystems, poised to evolve alongside the changing landscape of industries.

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